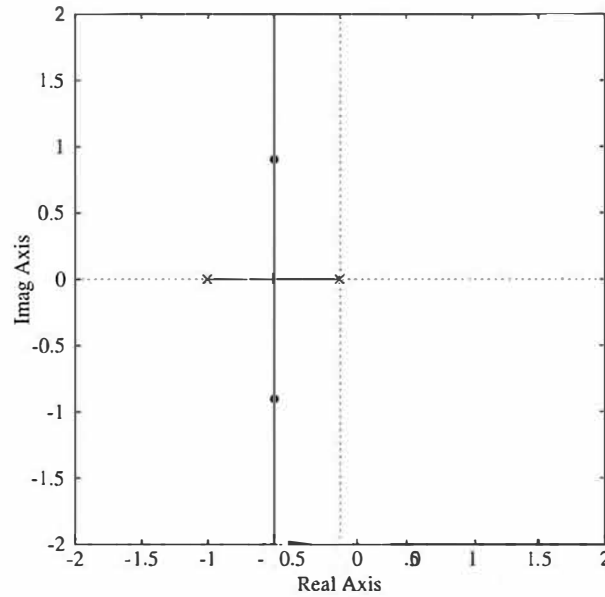


**Figure 7.4**  
Root locus for  
compensated antenna  
model



The locations of the roots with  $K = 1$  corresponding to a velocity constant of  $K_v = 1$  are marked by the dots computed by

$$p = \text{rlocus}(\text{sysol}, 1.0).$$

The natural frequency for the given pole locations is essentially  $\omega_n = 1$ , which corresponds to a rise time of  $t_r = 1.8$  sec. The indicated sampling period is thus  $T = t_r/10 = 0.18$ . A value of  $T = 0.2$  will be used for this example and a value of  $T = 1.0$  illustrated later to dramatize the effects of the choice of  $T$ . The compensation,  $D(s)$ , given by Eq. (7.4), has two first-order factors; the zero is at  $s = -0.1$ , and the pole is at  $s = -1$ . The pole-zero mapping technique requires that each singularity is mapped according to  $z = e^{sT}$ ; therefore, we take  $D(z)$  of the form

$$D(z) = K \frac{z - z_1}{z - p_1},$$

and place a zero at

$$z_1 = e^{(-0.1)(0.2)} = 0.9802,$$

and a pole at

$$p_1 = e^{(-1)(0.2)} = 0.8187.$$

To make the dc gain of  $D(z)$  and  $D(s)$  be identical, we require that

$$\begin{aligned} \text{dc gain} &= \lim_{z \rightarrow 1} D(z) = \lim_{s \rightarrow 0} D(s) = 1 \\ &= K \frac{1 - 0.9802}{1 - 0.8187} \end{aligned} \quad (7.5)$$

Solving for  $K$  we have

$$K = 9.15,$$

and the design of the discrete equivalent compensation has the transfer function

$$D(z) = 9.15 \frac{z - 0.9802}{z - 0.8187}. \quad (7.6)$$

To compute this result in MATLAB, the command is

$$\text{lead1d} = \text{c2d}(\text{lead1}, 0.2, \text{'matched'}).$$

### ◆ Example 7.3 Implementing the Controller

Give the difference equation that corresponds to the  $D(z)$  given by Eq. (7.6).

**Solution.** The transfer function is converted into a difference equation for implementation using the ideas developed in Chapter 4. Specifically, we first multiply top and bottom by  $z^{-1}$  to obtain

$$D(z) = \frac{U(z)}{E(z)} = 9.15 \frac{1 - 0.9802z^{-1}}{1 - 0.8187z^{-1}},$$

which can be restated as

$$(1 - 0.8187z^{-1})U(z) = 9.15(1 - 0.9802z^{-1})E(z).$$

The  $z$ -transform expression above is converted to the difference equation form by noting that  $z^{-1}$  represents a 1-cycle delay. Thus

$$u(k) = 0.8187u(k-1) + 9.15(e(k) - 0.9802e(k-1)).$$

This equation can be directly evaluated by a computer.

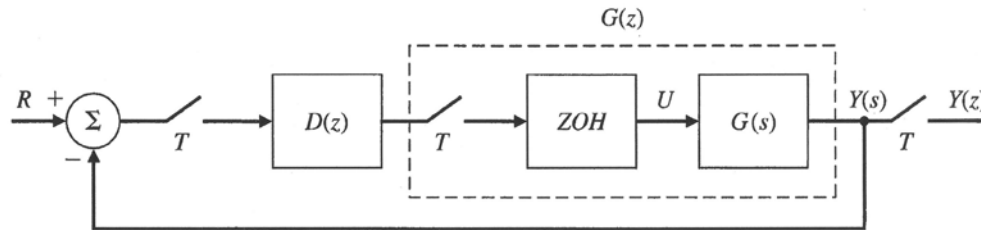
## 7.2.2 Evaluation of the Design

A description of a digital controller that is expected to satisfy the specifications for the antenna controller is now complete. A block diagram of the sampled-data system with discrete controller is given in Fig. 7.5. To analyze the behavior of this compensation, we first determine the  $z$ -transform of the continuous plant (Fig. 7.2) preceded by a zero-order hold (ZOH).

$$G(z) = \frac{z-1}{z} \mathcal{Z} \left\{ \frac{a}{s^2(s+a)} \right\}, \quad (7.7)$$

**Figure 7.5**

Block diagram of sampled-data system



which is

$$G(z) = \frac{z-1}{z} \mathcal{Z} \left\{ \frac{1}{s^2} - \frac{1}{as} + \frac{1}{a} \frac{1}{s+a} \right\}.$$

Using the tables in Appendix B, we find

$$\begin{aligned} G(z) &= \frac{z-1}{z} \left\{ \frac{Tz}{(z-1)^2} - \frac{z}{a(z-1)} + \frac{1}{a} \frac{z}{z-e^{-aT}} \right\} \\ &= \frac{Az+B}{a(z-1)(z-e^{-aT})}, \end{aligned}$$

where

$$A = e^{-aT} + aT - 1, \quad B = 1 - e^{-aT} - aTe^{-aT}.$$

For this example, with  $T = 0.2$  and  $a = 0.1$ , this evaluates to

$$G(z) = 0.00199 \frac{z + 0.9934}{(z-1)(z-0.9802)}. \quad (7.8)$$

Of course, this can be computed in MATLAB as the discrete model of the antenna by

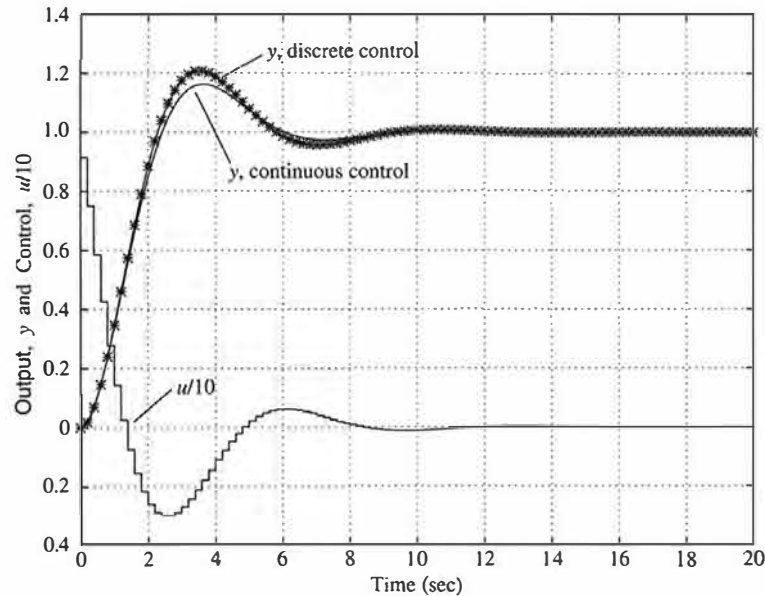
$$\text{antd} = \text{c2d}(\text{ant}, 0.2).$$

With the transfer function of the plant-plus-hold and the discrete controller, we can obtain the system difference equation and compute the step response, obviously most easily using a computer aided design package. The steps are

$$\begin{aligned} \text{sysold} &= \text{lead1d} * \text{antd} \\ \text{syscd} &= \text{connect}(\text{sysold}, [1 - 1]) \\ \text{step}(\text{syscd}). \end{aligned}$$

In this case, the step response of the system with the discrete controller is shown in Fig. 7.6. The figure confirms that the discrete controller will perform satisfactorily, albeit with somewhat increased overshoot. This simulation was carried out using the linear, discrete model of the system. As mentioned earlier, simulations can be embellished with the important nonlinearities such as friction and with

**Figure 7.6**  
Step response of the 5  
Hz controller



computation delays in order to assess their effects in addition to the effect of the discretization approximations.

#### ◆ Example 7.4 Antenna Servo with Slow Sampling

Repeat the antenna design with a sample rate of 1 Hz ( $T = 1$  sec); in this case the sample rate is approximately two samples per rise time.

**Solution.** Repeating the calculations as in Eq. (7.7) with  $T = 1$  sec results in

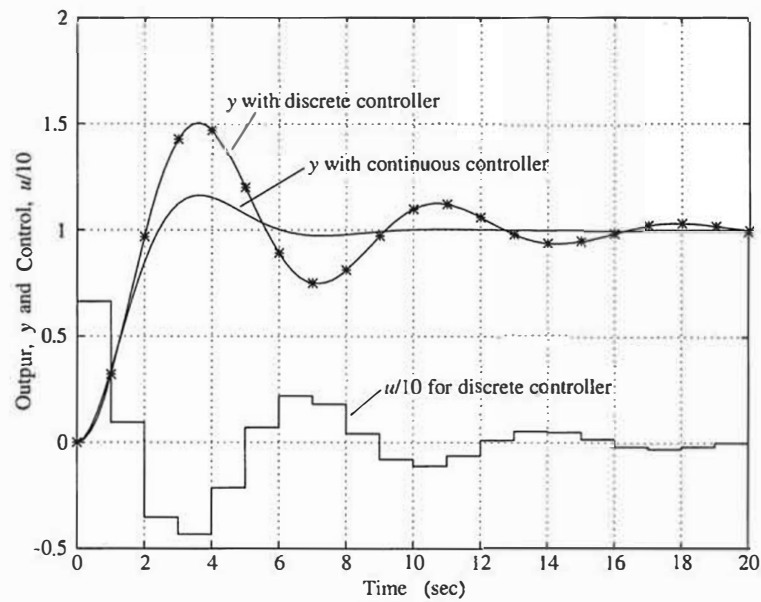
$$G(z) = 0.0484 \frac{z + 0.9672}{(z - 1)(z - 0.9048)}. \quad (7.9)$$

Furthermore, repeating the calculations that led to Eq. (7.6) but with  $T = 1$  sec, we obtain

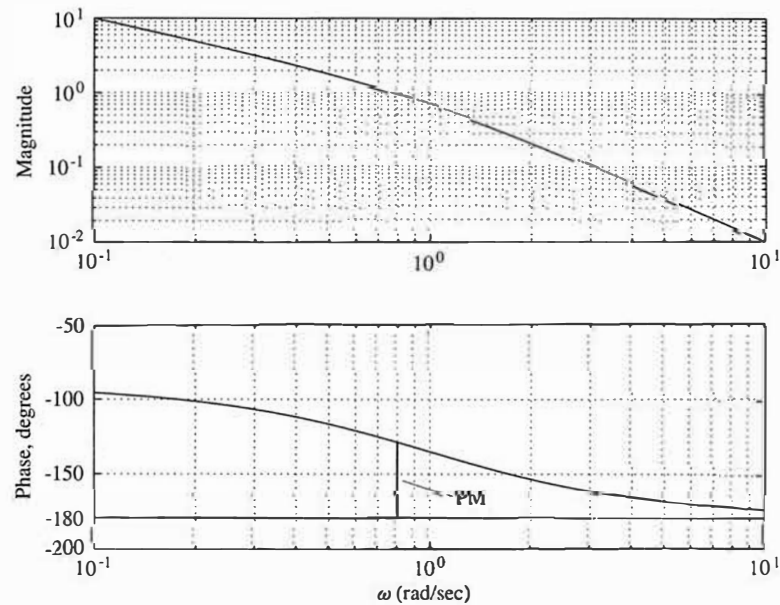
$$D(z) = 6.64 \frac{z - 0.9048}{z - 0.3679}. \quad (7.10)$$

A plot of the step response of the resulting system is shown in Fig. 7.7 and shows substantial degradation of the response as a result of the slow sampling. A partial explanation of the extra overshoot can be obtained by looking at the Bode plot of the continuous design, computed with `bode(sysol)` and plotted in Fig. 7.8. The designed phase margin in the continuous system is seen to be  $51.8^\circ$ . As was indicated in Chapter 4, the sample and hold can be roughly approximated by a delay of  $T/2$  sec. At the crossover frequency of  $\omega_{cp} = 0.8$  rad, and with sampling at  $T = 0.2$ , this corresponds only to  $\phi = \omega_{cp} T = 4.5^\circ$ . However, at  $T = 1.0$ , the sample-and-hold delay corresponds to  $\phi = 23^\circ$ . Thus the effective phase margin with a sample and hold is reduced to  $Pm = 51.8^\circ - 23^\circ = 28.8^\circ$ . With this small phase margin, the effective damping

**Figure 7.7**  
Step response of the  
1-Hz controller



**Figure 7.8**  
Bode plot of the  
continuous design for  
the antenna control



ratio is about 0.29 and the overshoot is expected to be about 0.4 rather than 0.16 as designed. The step response shows the actual  $M_p = 0.5$ , so most of the extra overshoot is explained by the sample-and-hold delay.

The examples presented here illustrate only a small selection of the alternatives for design by emulation. An immediate improvement would be expected if the continuous design were to include at the outset the characteristic  $T/2$  delay of the sample and zero-order hold. Other than this modification, the other algorithms for discrete equivalent design can be tried. These include the very simple Euler rectangular rules, the bilinear transformations, and the several hold-equivalent methods. The triangle hold equivalent appears to be especially promising.<sup>1</sup> There does not seem to be a dominant technique that is best for every case. The designer needs to explore alternatives based on the particular system, the required performance specifications and the practical constraints introduced by the technology to be used for implementation to guide the final choice. Here we now turn to consider the direct discrete design methods, beginning with design by use of the root locus in the  $z$ -plane.

### 7.3 Direct Design by Root Locus in the $z$ -Plane

The root locus introduced by W. Evans is based on graphical rules for plotting the roots of a polynomial as a parameter is varied. The most common root locus is a plot of the roots of a closed-loop characteristic polynomial in the  $s$ -plane as the loop gain is varied from 0 to  $\infty$ . In linear discrete systems also the dynamic performance is largely determined by the roots of the closed-loop characteristic polynomial, in this case a polynomial in  $z$  with stability represented by having all roots inside the unit circle. The consequences for direct digital design are that one can use Evans' root locus rules unchanged, but that the performance specifications must first be translated into the  $z$ -plane.

#### 7.3.1 $z$ -Plane Specifications

Figure 4.26 is a map of the unit disk in the  $z$ -plane on which is superimposed discrete system time responses that correspond to several typical  $z$ -plane pole locations. These can be used to make the translation of dynamic response performance specifications to a region of acceptable pole locations. For example, we have seen that rise time of a continuous second-order system is found to be inversely proportional to natural frequency as given by Eq. (7.1). Since poles in the  $s$ -plane are mapped to  $z = e^{sT}$ , the natural frequency in  $s$  maps to the angle of the pole in polar coordinates in the  $z$ -plane as  $\theta = \omega_d T$  where  $\omega_d = \sqrt{1 - \xi^2} \omega_n$ . Settling time is found to be inversely proportional to the magnitude of the real part of a pole in the  $s$ -plane ( $\sigma$ ) which maps to the radius of the pole in the  $z$ -plane as  $r = e^{-\sigma T}$ . The step response overshoot varies inversely with the damping

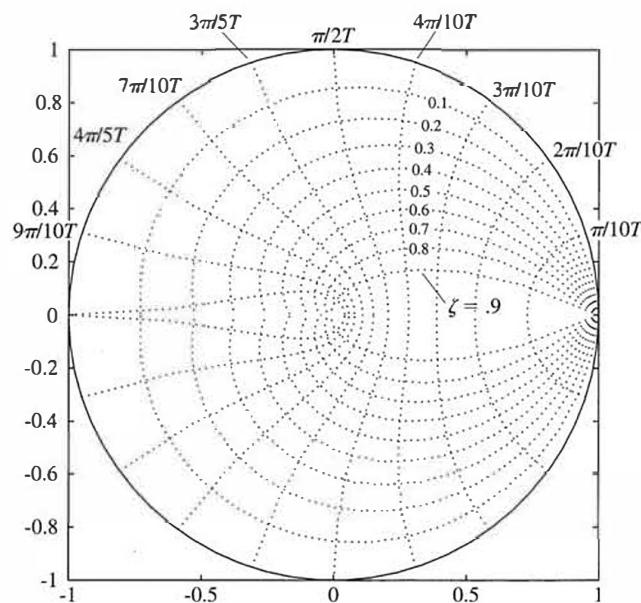
<sup>1</sup> In the MATLAB Control Toolbox function `c2d`, the triangle hold is called a first-order hold in recognition of the fact that it is a first-order hold although it is *noncausal*.

getting acceptable pole location in the z-plane

ratio. Under the  $s$ -to- $z$  mapping, lines of constant damping map into logarithmic spirals in the  $z$ -plane. With these guidelines, one can readily estimate the dynamic response parameters based on the pole-zero pattern for simple transfer functions and can derive useful guidelines for design of more complex systems. In summary, to get the specifications on acceptable pole locations in the  $z$ -plane

- Estimate the desired  $\omega_n$ ,  $\zeta$ , and  $M_p$  from the continuous-time response specifications. Compute  $\sigma = \zeta \omega_n$ .
- Compute the radius  $r = e^{-\sigma T}$ .
- Obtain a plot of the  $z$ -plane showing lines of fixed damping and  $\omega_n$ . The MATLAB command `zgrid` will do this, plotting  $\zeta$  in steps of 0.1 from 0.1 to 0.9 and  $\omega_n = N\pi/10T$  for integer  $N$  from 1 to 10. An example is shown in Fig. 7.9. The command `axis equal` will cause the unit circle to be plotted as a circle and the command `axis([-1 1 0 1])` will cause only the upper half of the circle to be plotted.
- Mark the region of acceptable closed-loop pole locations on the plane.

**Figure 7.9**  
Lines of constant damping and natural frequency in the  $z$ -plane



◆ **Example 7.5** *Z-Plane Specifications*

Indicate on a  $z$ -plane map the region of acceptable closed-loop poles for the antenna design of Example 7.2.

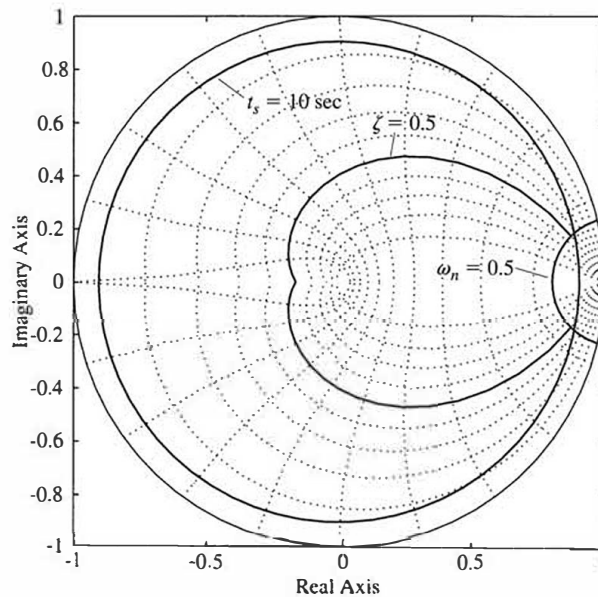
**Solution.** The given specifications are that the system is to have a damping ratio of  $\zeta \geq 0.5$ , natural frequency of  $\omega_n \geq 1$ , and the real-parts of the roots are to be greater than 0.5. The standard grid of the  $z$ -plane shows the curve corresponding to  $\zeta = 0.5$ . With the requirement that the roots correspond to a natural frequency greater than  $\omega_n = 1$ , we need a plot on the standard grid corresponding to  $N = 10T\omega_n/\pi = 2/\pi \simeq 0.64$ . The last requirement means that the roots in the  $z$ -plane must be inside a circle of radius  $r \leq e^{-0.5T} = 0.9048$ . The curves corresponding to these criteria are marked in Fig. 7.10.

The specification of steady-state error also follows the continuous case but transferred to the  $z$ -plane when the controller is implemented in a computer and represented by its discrete transfer function  $D(z)$ . The discrete transfer function of the plant is given by

$$G(z) = (1 - z^{-1})\mathcal{Z}\left\{\frac{G(s)}{s}\right\}. \quad (7.11)$$

The closed-loop system can now be represented in a purely discrete manner. The discrete transfer functions of the controller,  $D(z)$ , and the plant,  $G(z)$ , are

**Figure 7.10**  
Plot of acceptable pole locations in the  $z$ -plane





combined as before according to Fig. 7.5, where it is now understood that the reference  $r$  and the disturbance  $w$  are sampled versions of their continuous counterparts. Proceeding as we did for the continuous system, suppose the input  $r$  is a step,  $r(k) = 1(k)$ , and the disturbance  $w$  is zero. The transform of the error is computed using the same block-diagram reduction tools that apply for continuous systems represented by their Laplace transforms, except that now we use  $D(z)$  and  $G(z)$ . Doing this yields the transform of the error

$$\begin{aligned} E(z) &= \frac{R(z)}{1 + D(z)G(z)} \\ &= \frac{z}{z-1} \frac{1}{1 + D(z)G(z)}. \end{aligned}$$

discrete time final value

The final value of  $e(k)$ , if the closed loop system is stable with all roots of  $1 + DG = 0$  inside the unit circle, is, by Eq. (4.115)

$$\begin{aligned} e(\infty) &= \lim_{z \rightarrow 1} (z-1) \frac{z}{z-1} \frac{1}{1 + D(z)G(z)} \\ &= \frac{1}{1 + D(1)G(1)} \\ &= \frac{1}{1 + K_p}. \end{aligned} \quad (7.12)$$

discrete system type

Thus,  $D(1)G(1)$  is the position error constant,  $K_p$ , of the Type 0 system in discrete time if the limit in Eq. (7.12) is finite. If  $DG$  has a pole at  $z = 1$ , then the error given by Eq. (7.12) is zero. Suppose there is a single pole at  $z = 1$ . Then we have a Type 1 system and we can compute the error to a unit ramp input,  $r(kT) = kT1(kT)$  as

$$E(z) = \frac{Tz}{(z-1)^2} \frac{1}{1 + D(z)G(z)}.$$

Now the steady-state error is

$$\begin{aligned} e(\infty) &= \lim_{z \rightarrow 1} (z-1) \frac{Tz}{(z-1)^2} \frac{1}{1 + DG} \\ &= \lim_{z \rightarrow 1} \frac{Tz}{(z-1)(1 + DG)} \\ &\triangleq \frac{1}{K_v}. \end{aligned} \quad (7.13)$$

Thus the **velocity constant** of a Type 1 discrete system with unity feedback (as shown in Fig. 7.5) is

$$K_v = \lim_{z \rightarrow 1} \frac{(z-1)(1 + D(z)G(z))}{Tz},$$

which simplifies to

$$K_v = \lim_{z \rightarrow 1} \frac{(z-1)D(z)G(z)}{Tz}. \quad (7.14)$$

Although it appears from Eq. (7.14) that  $K_v$  is inversely proportional to the sample period, this is not the case if comparing for the same  $G(s)$ . The reason is that the transfer function of  $G(z)$  computed from Eq. (7.11) is typically proportional to the sample period. This proportionality is exact for the very simple case where  $G(s) = 1/s$ , as can be seen by using Eq. (7.11) and inspecting Entry 4 in Appendix B.2. For systems with a finite  $K_v$  and fast sample rates, this proportionality will be approximately correct. The result of this proportionality is that the dc gain of a continuous plant alone preceded by a ZOH is essentially the same as that of the continuous plant.

#### \*Truxal's Rule, Discrete Case

Because systems of Type 1 occur frequently, it is useful to observe that the value of  $K_v$  is fixed by the *closed-loop* poles and zeros by a relation given, for the continuous case, by Truxal (1955). Suppose the overall transfer function  $Y/R$  is  $H(z)$ , and that  $H(z)$  has poles  $p_i$  and zeros  $z_i$ . Then we can write

$$H(z) = K \frac{(z-z_1)(z-z_2) \cdots (z-z_n)}{(z-p_1)(z-p_2) \cdots (z-p_n)}. \quad (7.15)$$

Now suppose that  $H(z)$  is the closed-loop transfer function that results from a Type 1 system, which implies that the steady-state error of this system to a step is zero and requires that

$$H(1) = 1. \quad (7.16)$$

Furthermore, by definition we can express the error to a ramp as

$$\begin{aligned} E(z) &= \frac{R(z)(1-H(z))}{Tz} \\ &= \frac{Tz}{(z-1)^2} (1-H(z)), \end{aligned}$$

and the final value of this error is given by

$$e(\infty) = \lim_{z \rightarrow 1} (z-1) \frac{Tz}{(z-1)^2} (1-H(z)) \triangleq \frac{1}{K_v};$$

therefore (omitting a factor of  $z$  in the numerator, which makes no difference in the result)

$$\frac{1}{TK_v} = \lim_{z \rightarrow 1} \frac{1-H(z)}{z-1}. \quad (7.17)$$

Because of Eq. (7.16), the limit in Eq. (7.17) is indeterminate, and so we can use L'Hôpital's rule

$$\begin{aligned}\frac{1}{TK_v} &= \lim_{z \rightarrow 1} \frac{(d/dz)(1 - H(z))}{(d/dz)(z - 1)} \\ &= \lim_{z \rightarrow 1} \left\{ -\frac{dH(z)}{dz} \right\}.\end{aligned}$$

However, note that by using Eq. (7.16) again, at  $z = 1$ , we have

$$\frac{d}{dz} \ln H(z) = \frac{1}{H} \frac{d}{dz} H(z) = \frac{d}{dz} H(z),$$

so that

$$\begin{aligned}\frac{1}{TK_v} &= \lim_{z \rightarrow 1} -\frac{d}{dz} \ln H(z) \\ &= \lim_{z \rightarrow 1} -\frac{d}{dz} \left\{ \ln K \frac{\prod (z - z_i)}{\prod (z - p_i)} \right\} \\ &= \lim_{z \rightarrow 1} -\frac{d}{dz} \left\{ \sum \ln(z - z_i) - \sum \ln(z - p_i) + \ln K \right\} \\ &= \lim_{z \rightarrow 1} \left\{ \sum \frac{1}{z - p_i} - \sum \frac{1}{z - z_i} \right\} \\ &= \sum_{i=1}^n \frac{1}{1 - p_i} - \sum_{i=1}^n \frac{1}{1 - z_i}.\end{aligned}$$

We note especially that the farther the *poles* of the closed-loop system are from  $z = 1$ , the larger the velocity constant and the smaller the errors. Similarly,  $K_v$  can be increased and the errors decreased by *zeros close* to  $z = 1$ . From the results of Chapter 4 on dynamic response, we recall that a zero close to  $z = 1$  usually yields large overshoot and poor dynamic response. Thus is expressed one of the classic trade-off situations: We must balance small steady-state errors against good transient response.

### 7.3.2 The Discrete Root Locus

The root locus is the locus of points where roots of a characteristic equation can be found as some real parameter varies from zero to large values. From Fig. 7.5 and block-diagram analysis, the characteristic equation of the single-loop system is

$$1 + D(z)G(z) = 0. \quad (7.18)$$

The significant thing about Eq. (7.18) is that this is exactly the same equation as that found for the  $s$ -plane root locus. The implication is that the *mechanics* of drawing the root loci are exactly the same in the  $z$ -plane as in the  $s$ -plane; the rules for the locus to be on the real axis, for asymptote construction, and

for arrival/departure angles are all unchanged from those developed for the  $s$ -plane and reviewed in Chapter 2. As mentioned earlier, the difference lies in the interpretation of the results because the pole locations in the  $z$ -plane mean different things than pole locations in the  $s$ -plane when we come to interpret system stability and dynamic response.

### ◆ Example 7.6 Discrete Root Locus Design

Design the antenna system for the slow sampling case with  $T = 1$  sec. using the discrete root locus.

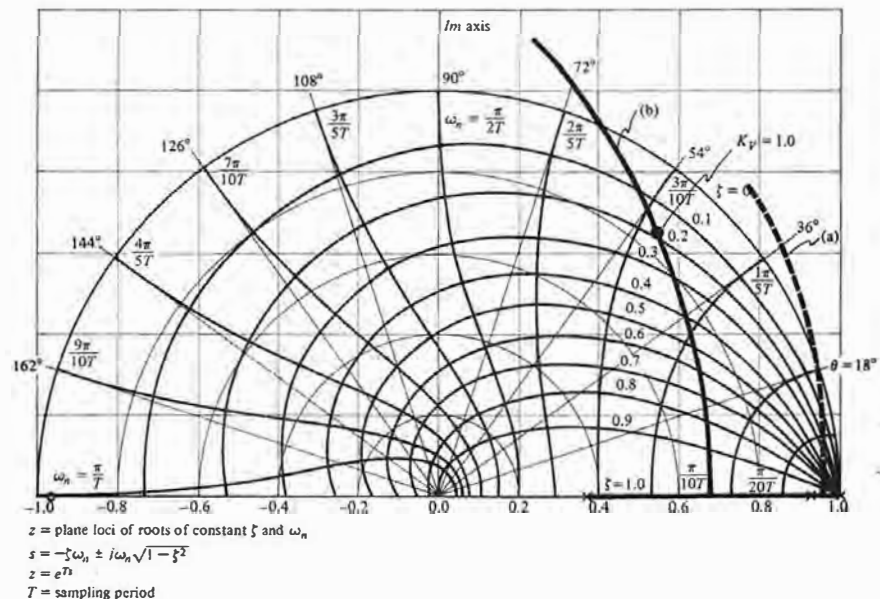
**Solution.** The exact discrete model of the plant plus hold is given by the  $G(z)$  in Eq. (7.9). If the controller consisted simply of a proportional gain [ $D(z) = K$ ], the locus of roots versus  $K$  can be found by solving the characteristic equation

$$1 + 0.0484K \frac{z + 0.9672}{(z - 1)(z - 0.9048)} = 0$$

for many values of  $K$ . The result computed by `rlocus(antd)` is shown in Fig. 7.11 as the dashed arc marked (a). From study of the root locus we should remember that this locus, with two poles and one zero, is a circle centered at the zero ( $z = -0.9672$ ) and breaking away from the real axis between the two real poles.

From the root locus of the uncompensated system (Fig. 7.11(a)) it is clear that some dynamic compensation is required if we are to get satisfactory response from this system. The

**Figure 7.11**  
Root loci for antenna design: (a) Uncompensated system; (b) Locus with  $D(z)$  having the poles and zeros of Eq. (7.10)



radius of the roots never gets less than 0.95, preventing the  $t_s$  specification from being met. The system goes unstable at  $K \cong 2.25$  [where  $K_v = K$ , as can be verified by using Eq. (7.14)], which means that there is no stable value of gain that meets the steady-state error specification with this compensation.

If we cancel the plant pole at 0.9048 with a zero and add a pole at 0.3679, we are using the lead compensation of Eq. (7.10). The root locus for this control versus the gain  $K$  [ $K$  was equal to 6.64 in Eq. (7.10)] computed with `rlocus(sysold)` is also sketched in Fig. 7.11 as the solid curve (b). The points,  $p$ , where  $K = 6.64$  are computed with `p = rlocus(sysold, 6.64)` and marked by dots. We can see that a damping ratio of about 0.2 is to be expected, as we have previously seen from the step response of Fig. 7.7. This gain, however, does result in the specified value of  $K_v = 1$  because this criterion was used in arriving at Eq. (7.10). The locus shows that increasing the gain,  $K$ , would lower the damping ratio still further. Better damping could be achieved by decreasing the gain, but then the criterion of steady-state error would be violated. It is therefore clear that this choice of compensation pole and zero cannot meet the specifications.

A better choice of compensation can be expected if we transform the specifications into the  $z$ -plane and select the compensation so that the closed loop roots meet those values. The original specifications were  $K_v \geq 1$ ,  $t_s \leq 10$  sec, and  $\zeta \geq 0.5$ . If we transform the specifications to the  $z$ -plane we compute that the  $t_s$  specification requires that the roots be inside the radius  $r = e^{-0.5} = 0.61$ , and the overshoot requires that the roots are inside the  $\zeta = 0.5$  spiral. The requirement that  $K_v \geq 1$  applies in either plane but is computed by Eq. (7.14) for the  $z$ -plane.

It is typically advantageous to use the design obtained using emulation and to modify it using discrete design methods so that it is acceptable. The problem with the emulation-based design is that the damping is too low at the mandated gain, a situation that is typically remedied by adding more lead in the compensation. More lead is obtained in the  $s$ -plane by increasing the separation between the compensation's pole and zero; and the same holds true in the  $z$ -plane. Therefore, for a first try, let's keep the zero where it is (canceling the plant pole) and move the compensation pole to the left until the roots and  $K_v$  are acceptable. After a few trials, we find that there is no pole location that satisfies all the requirements! Although moving the pole to the left of  $z \cong 0$  will produce acceptable  $z$ -plane pole locations, the gain  $K_v$  is not sufficiently high to meet the criterion for steady-state error. The only way to raise  $K_v$  and to meet the requirements for damping and settling time is to move the zero to the left also.

After some trial and error, we see that

$$D(z) = 6 \frac{z - 0.80}{z - 0.05} \quad (7.19)$$

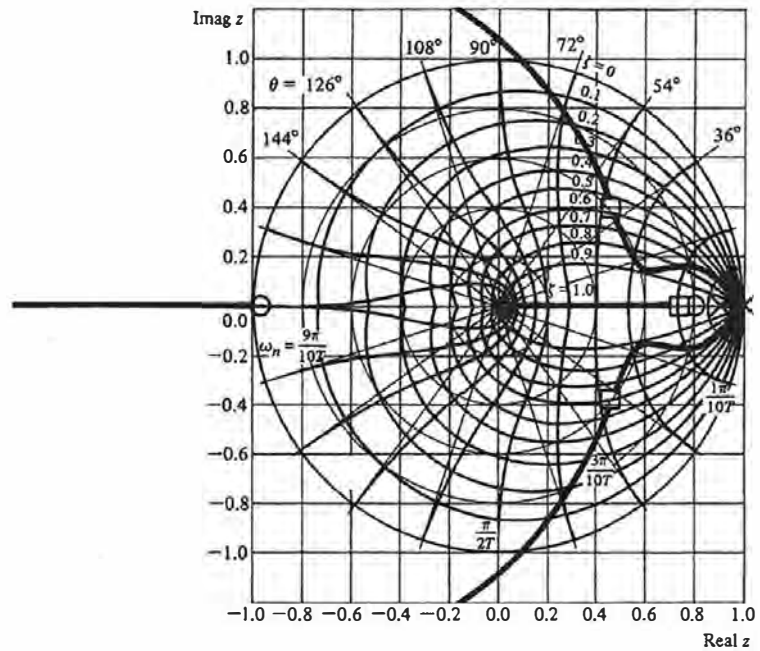
meets the required  $z$ -plane constraints for the complex roots and has a  $K_v = 1.26$ . The root locus for Eq. (7.19) is shown in Fig. 7.12(a), and the roots corresponding to  $K = 6$  are marked by squares. The fact that all requirements seem to be met is encouraging, but there is an additional real root at  $z = 0.74$  and a zero at  $z = 0.8$ , which may degrade the actual response from that expected if it were a second-order system. The actual time history is shown in Fig. 7.12(b). It shows that the overshoot is 29% and the settling time is 15 sec. Therefore, further iteration is required to improve the damping and to prevent the real root from slowing down the response.

A compensation that achieves the desired result is

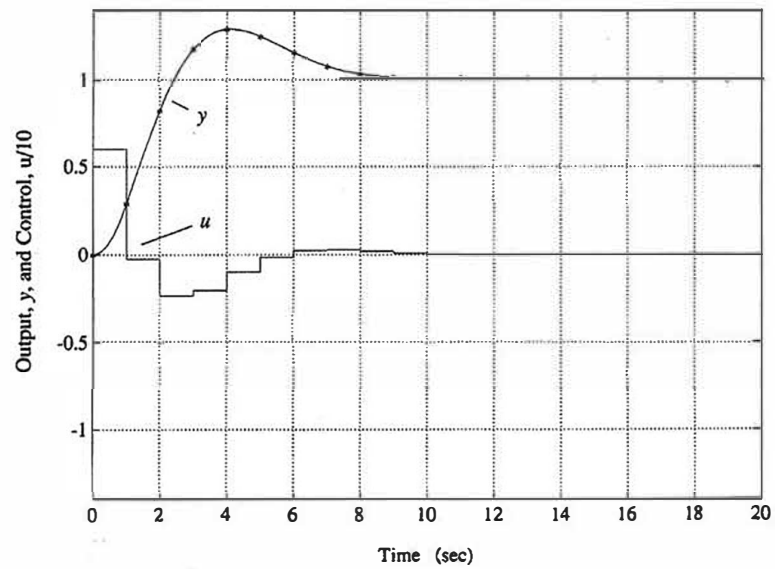
$$D(z) = 13 \frac{z - 0.88}{z + 0.5} \quad (7.20)$$

**Figure 7.12**

Antenna design with  $D(z)$  given by Eq. (7.19):  
(a) root locus, (b) step response



(a)



(b)

The damping and radius of the complex roots substantially exceed the specified limits, and  $K_v = 1.04$ . Although the real root is slower than the previous design, it is very close to a zero that attenuates its contribution to the response. The root locus for all  $K$ 's is shown in Fig. 7.13(a) and the time response for  $K = 13$  in Fig. 7.13(b).

Note that the pole of Eq. (7.20) is on the negative real  $z$ -plane axis. In general, placement of poles on the negative real axis should be done with some caution. In this case, however, no adverse effects resulted because all roots were in well-damped locations. As an example of what could happen, consider the compensation

$$D(z) = 9 \frac{(z - 0.8)}{(z + 0.8)}. \quad (7.21)$$

The root locus versus  $K$  and the step response are shown in Fig. 7.14. All roots are real with one root at  $z = -0.59$ . But this negative real axis root has  $\zeta = 0.2$  and represents a damped sinusoid with frequency of  $\omega_s/2$ . The output has very low overshoot, comes very close to meeting the settling time specification, and has  $K_v = 1$ ; however, the control,  $u$ , has large oscillations with a damping and frequency consistent with the negative real root. This indicates that there are "hidden oscillations" or "intersample ripple" in the output that are only apparent by computing the continuous plant output between sample points as is done in Fig. 7.14. The computation of the intersample behavior was carried out by computing it at a much higher sample rate than the digital controller, taking care that the control value was constant throughout the controller sample period. The MATLAB function `ripple`, included in the Digital Control Toolbox, has been written to do these calculations. Note that if only the output at the sample points had been determined, the system would appear to have very good response. This design uses much more control effort than that shown in Fig. 7.13, a fact that is usually very undesirable. So we see that a compensation pole in a lightly damped location on the negative real axis could lead to a poorly damped system pole and undesirable performance.

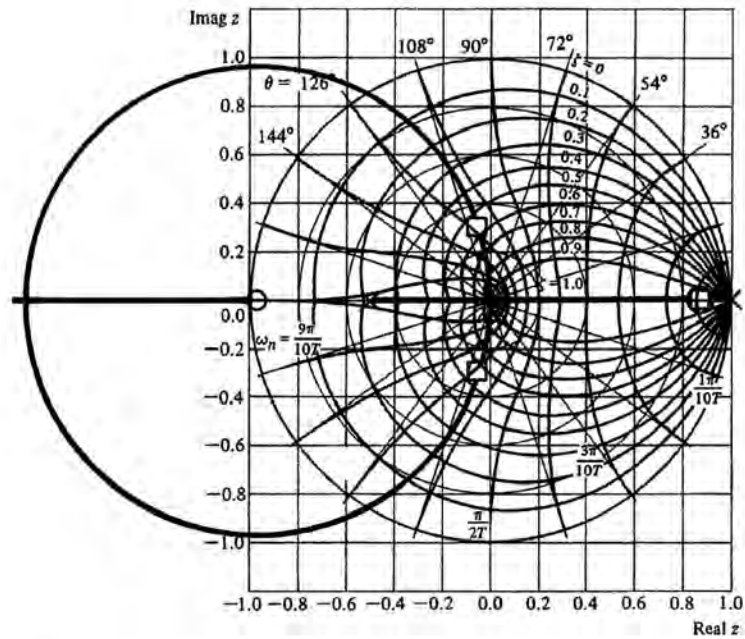
`ripple`

In the design examples to this point, the computed output time histories have assumed that the control,  $u(k)$ , was available from the computer at the sample instant. However, in a real system this is not always true. In the control implementation example in Table 3.1, we see that some time must pass between the sample of  $y(k)$  and the output of  $u(k)$  for the computer to calculate the value of  $u(k)$ . This time delay is called **latency** and usually can be kept to a small fraction of the sample period with good programming and computer design. Its effect on performance can be evaluated precisely using the transform analysis of Section 4.4.2, the state-space analysis of Section 4.3.4, or the frequency response. The designer can usually determine the expected delay and account for it in the design. However, if not taken into account, the results can be serious as can be seen by an analysis using the root locus.

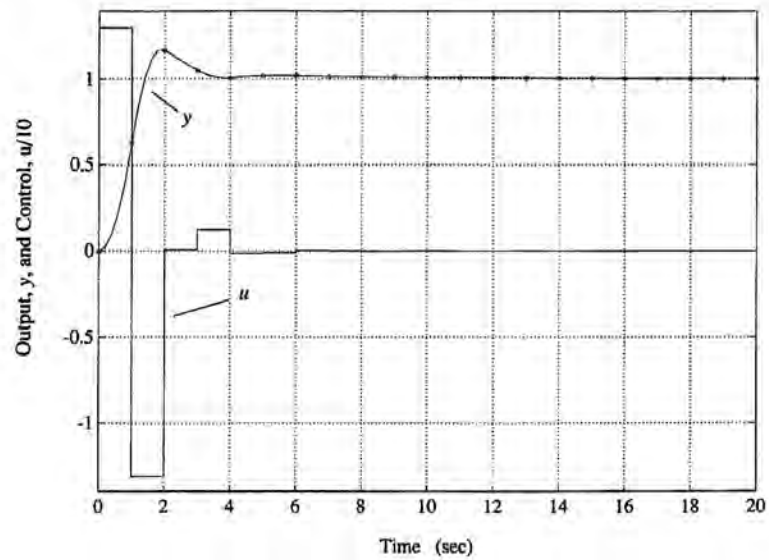
Because a one-cycle delay has a  $z$ -transform of  $z^{-1}$ , the effect of a full-cycle delay can be analyzed by adding  $z^{-1}$  to the numerator of the controller representation. This will result in an additional pole at the origin of the  $z$ -plane. If there is a delay of two cycles, two poles will be added to the  $z$ -plane origin, and so on.

**Figure 7.13**

Antenna design with  $D(z)$  given by Eq. (7.20):  
(a) root locus, (b) step response



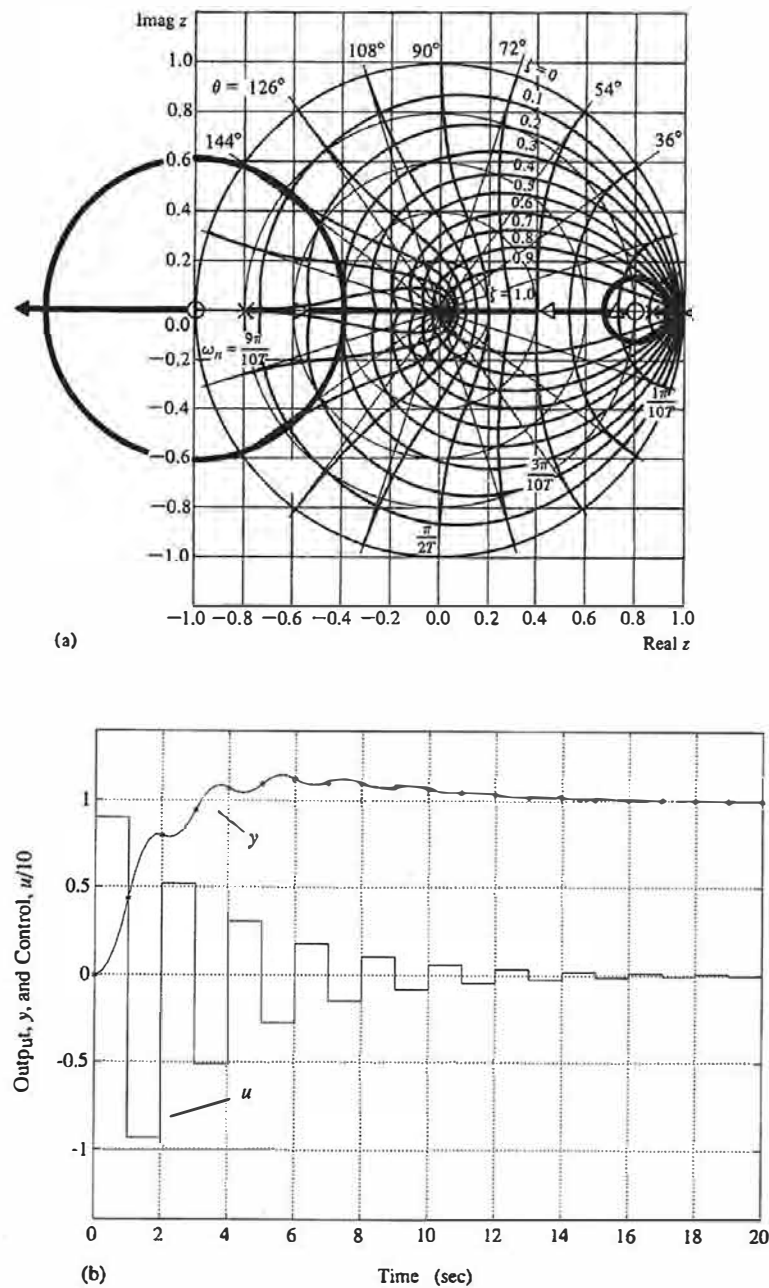
(a)



(b)



**Figure 7.14**  
Antenna design with  
 $D(z)$  given by Eq. (7.21):  
(a) root locus, (b) step  
response



◆ **Example 7.7** *Effect of Unexpected Delay*

Add one cycle delay to the compensation of Eq. (7.21) and plot the resulting root locus and step response.

**Solution.** The new controller representation is

$$D(z) = 13 \frac{z - 0.88}{z(z + 0.5)}. \quad (7.22)$$

The root locus and time response are shown in Fig. 7.15, which are both substantially changed from the same controller without the delay as shown in Fig. 7.13. The only difference is the new pole at  $z = 0$ . The severity of the one-cycle delay is due to the fact that this controller is operating at a very slow sample rate (six times the closed loop bandwidth). This sensitivity to delays is one of many reasons why one would prefer to avoid sampling at this slow a rate.

## 7.4 Frequency Response Methods

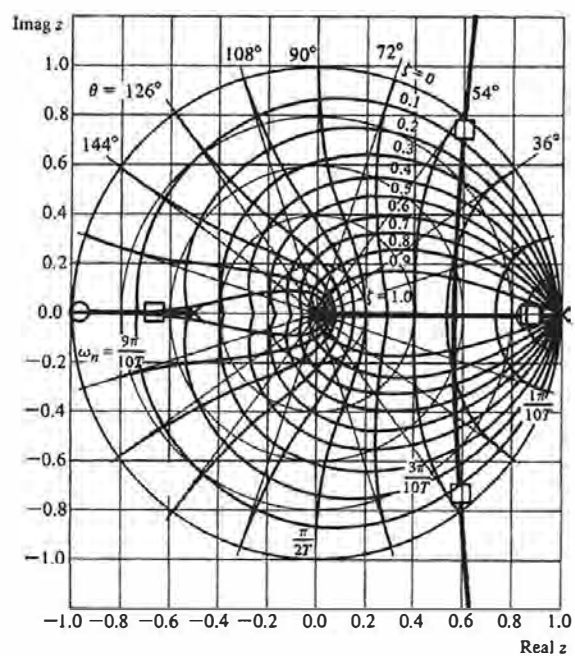
The frequency response methods for continuous control system design were developed from the original work of Bode (1945) on feedback-amplifier techniques. Their attractiveness for design of continuous linear feedback systems depends on several ideas.

1. The gain and phase curves for a rational transfer function can be easily plotted *by hand*.
2. If a physical realization of the system is available, the frequency response can be measured experimentally without the necessity of having a mathematical model at all.
3. Nyquist's stability criterion can be applied, and dynamic response specifications can be readily interpreted in terms of gain and phase margins, which are easily seen on the plot of log gain and phase-versus-log frequency.
4. The system error constants, mainly  $K_p$  or  $K_v$ , can be read directly from the low-frequency asymptote of the gain plot.
5. The corrections to the gain and phase curves (and thus the corrections in the gain and phase margins) introduced by a trial pole or zero of a compensator can be quickly and easily computed, *using the gain curve alone*.
6. The effect of pole, zero, or gain changes of a compensator on the speed of response (which is proportional to the crossover frequency) can be quickly and easily determined *using the gain curve alone*.

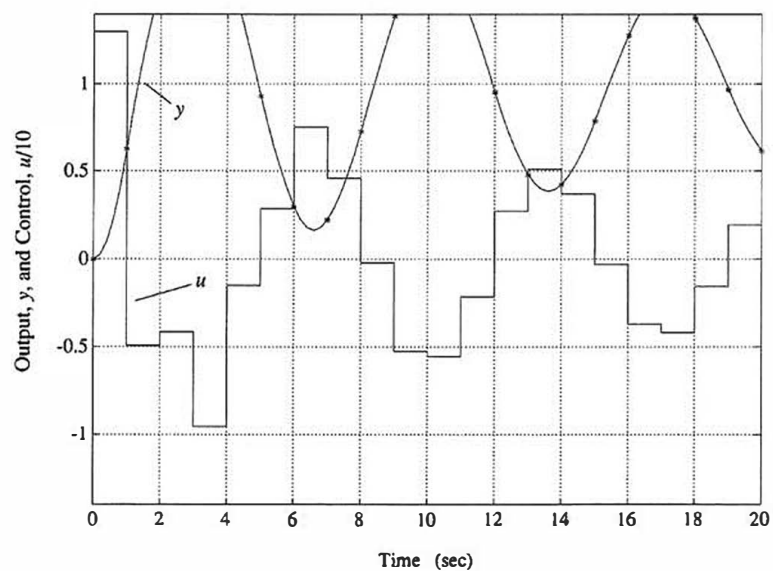
Use of the frequency response in the design of continuous systems has been reviewed in Chapter 2 and the idea of discrete frequency responses has

**Figure 7.15**

One-cycle-delay antenna design with  $D(z)$  given by Eq. (7.22): (a) root locus, (b) step response



(a)



(b)

been introduced in Chapter 4. In order to apply these concepts to the design of digital controls, the basic results on stability and performance must be translated to the discrete domain. The concepts are the same as for continuous systems, but plots of the magnitude and phase of a discrete transfer function,  $H(z)$ , are accomplished by letting  $z$  take on values around the unit circle,  $z = e^{j\omega T}$ , that is,

$$\begin{aligned}\text{magnitude} &= |H(z)|_{e^{j\omega T}}, \\ \text{phase} &= \angle H(z)|_{e^{j\omega T}}.\end{aligned}$$

### ◆ Example 7.8 Discrete Bode Plot

Plot the discrete frequency response corresponding to the plant transfer function

$$G(s) = \frac{1}{s(s+1)} \quad (7.23)$$

sampling with a zero order hold at  $T = 0.2, 1$ , and  $2$  seconds and compare with the continuous response.

**Solution.** The discrete transfer functions for the specified sampling periods are computed with `c2d.m` as

```
sysc = tf([1],[1 1 0])
sysd1=c2d(sysc,0.2)
sysd2 = c2d(sysc,1)
sysd3=c2d(sysc,2)
```

with transfer functions

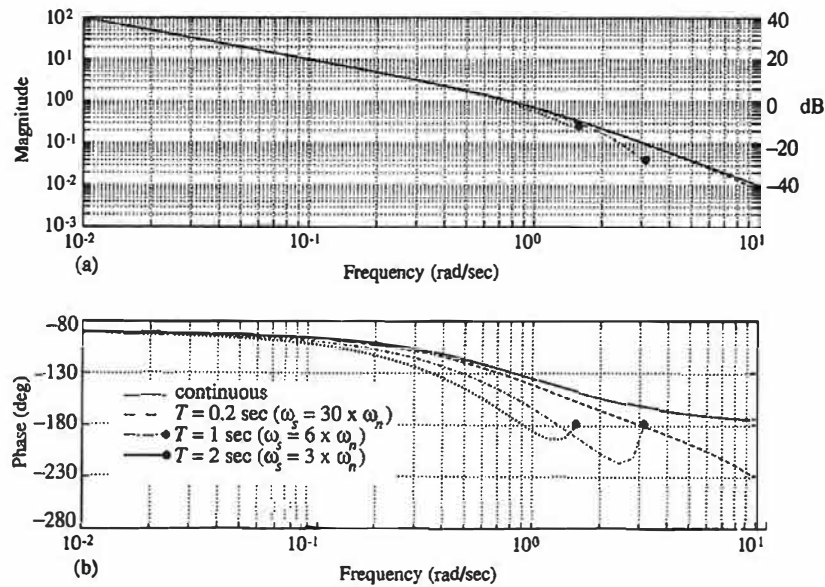
$$\begin{aligned}G_1(z) &= 0.0187 \frac{z + 0.9355}{(z-1)(z-0.8187)} \quad \text{for } T = 0.2 \text{ sec} \\ G_2(z) &= 0.368 \frac{z + 0.718}{(z-1)(z-0.368)} \quad \text{for } T = 1 \text{ sec} \\ G_3(z) &= 1.135 \frac{z + 0.523}{(z-1)(z-0.135)} \quad \text{for } T = 2 \text{ sec}.\end{aligned} \quad (7.24)$$

The frequency responses of Eq. (7.23) and Eq. (7.24) are plotted in Fig. 7.16 using the statement

```
bode(sysc,'-',sysd1,'-.',sysd2,':',sysd3,'-').
```

It is clear that the curves for the discrete systems are nearly coincident with the continuous plot for low frequencies but deviate substantially as the frequency approaches  $\pi/T$  in each case. In particular, the amplitude plots do not approach the simple asymptotes used in the hand-plotting procedures developed by Bode, and his theorem relating the phase to the derivative of the magnitude curve on

**Figure 7.16**  
Frequency responses of  
continuous and discrete  
transfer functions



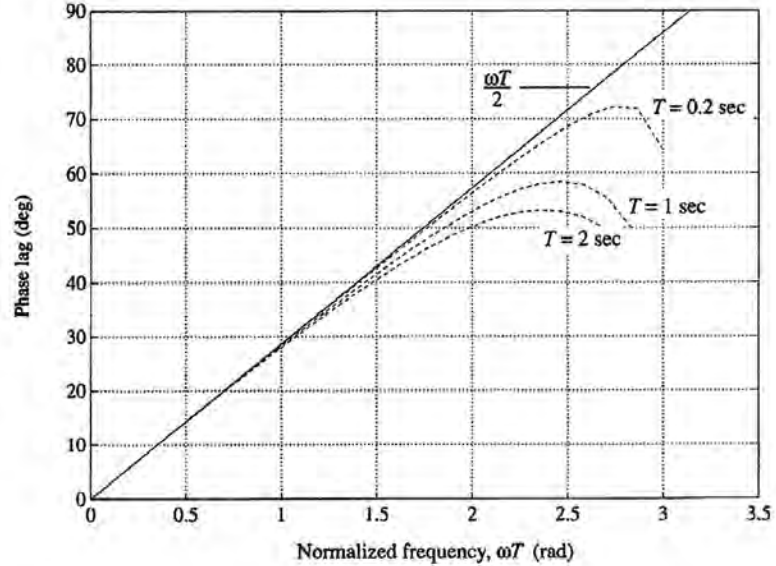
a log-log plot does not apply. The primary effect of sampling is to cause an additional phase lag. Fig. 7.17 shows this additional phase lag by plotting the phase difference,  $\Delta\phi$ , between the continuous case and the discrete cases. The approximation to the discrete phase lag given by

$$\Delta\phi = \frac{\omega T}{2} \quad (7.25)$$

is also shown and demonstrates the accuracy of this approximation for sample rates up to  $\omega T = \pi/2$ , which corresponds to frequencies up to 1/4 the sample rate. Crossover frequencies (where magnitude = 1) for designs will almost always be lower than 1/4 the sample rate; therefore, one can obtain a good estimate of the phase margin if a sample and hold is introduced into a continuous design by simply subtracting the factor  $\omega T/2$  from the phase of the continuous design's phase margin.

The inability to use the standard plotting guidelines detracts from the ease with which a designer can predict the effect of pole and zero changes on the frequency response. Therefore, points 1, 5, and 6 above are less true for discrete frequency-response design using the z-transform than they are for continuous systems and we are more dependent on computer aids in the discrete case. With some care in the interpretations however, points 2, 3, and 4 are essentially unchanged. All these points will be discussed further in this chapter as they pertain to design using the *discrete* frequency response. We begin with the discrete form of the **Nyquist stability criterion** and follow with a discussion of specifications

**Figure 7.17**  
Phase lag due to  
sampling



of performance and stability robustness as expressed in the frequency domain before we introduce the design techniques directly.

### 7.4.1 Nyquist Stability Criterion

For continuous systems, the Nyquist stability criterion seeks to determine whether there are any zeros of the closed-loop characteristic equation

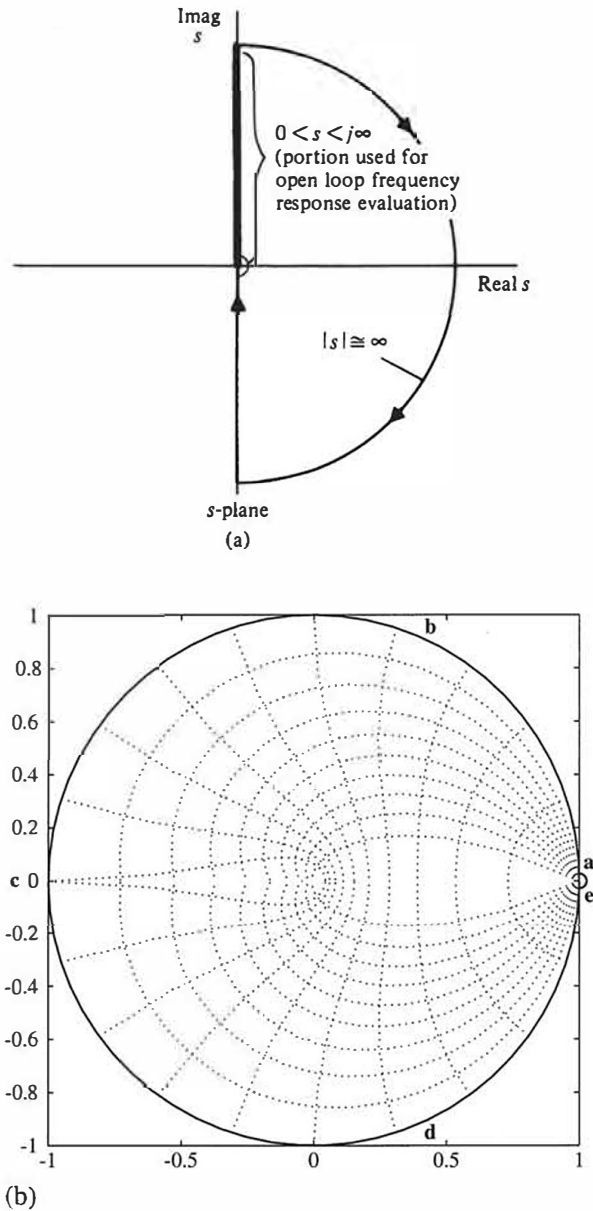
$$1 + K D(s)G(s) = 0 \quad (7.26)$$

in the right half-plane. The method establishes stability by determining the excess of zeros over poles of the characteristic equation in the right-half plane by plotting  $K D(s)G(s)$  for  $s$  along the  $\mathcal{D}$  contour that encloses the entire right-hand side (unstable region) of the  $s$ -plane as sketched in Fig. 7.18(a).

It is assumed that the designer *knows* the number of (unstable) poles that are inside the contour and from the plot can then determine the number of zeros of Eq. (7.26) in the unstable region that is the same as the number of closed-loop system unstable poles. The entire contour evaluation is fixed by examining  $K D(s)G(s)$  over  $s = j\omega$  for  $0 \leq \omega < \infty$ , which is the frequency-response evaluation of the open-loop system. For experimental data, the plot is to be made for  $\omega_{low} \leq \omega \leq \omega_{high}$ , where  $\omega_{low}$  is small enough to allow the low-frequency behavior to be decided (usually the gain is very high at  $\omega_{low}$  and the phase is approaching a fixed multiple of  $90^\circ$ ), and where  $\omega_{high}$  is taken to be high enough that it is known that the magnitude is much less than 1 for all higher

**Figure 7.18**

Contours used for  
Nyquist stability criterion:  
(a) In the  $s$ -plane; (b) In  
the  $z$ -plane



frequencies. Fig. 7.18(a) shows the full  $\mathcal{D}$  contour and the portion of the contour for  $\omega_{low} \leq \omega \leq \omega_{high}$ . The indentation near  $\omega = 0$  excludes the (known) poles of  $KDG$  at  $s = 0$  from the unstable region; the map of this small semicircle is done analytically by letting  $s = r^{j\phi}$  for  $r \ll 1$ ,  $-\frac{\pi}{2} \leq \phi \leq \frac{\pi}{2}$ .

The specific statement of the Nyquist stability criterion for continuous systems is

$$Z = P + N, \quad (7.27)$$

where

- $Z$  = the number of unstable zeros of Eq. (7.26) (that are unstable closed-loop poles). For stability,  $Z = 0$ .
- $P$  = the number of unstable (open-loop) poles of  $KD(s)G(s)$ .
- $N$  = the net number of encirclements of the  $-1$  point for the contour evaluation of  $KD(s)G(s)$  in the *same direction* as that taken by  $s$  along  $\mathcal{D}$  as shown in Fig 7.18(a). Usually  $s$  is taken clockwise around  $\mathcal{D}$  and therefore clockwise encirclements are taken as positive.

For the common case of a stable open-loop system ( $P = 0$ ) the closed-loop system is stable if and only if the contour evaluation of  $KD(s)G(s)$  does not encircle the  $-1$  point. For unstable open-loop systems, the closed-loop system is stable if and only if the contour evaluation encircles the  $-1$  point counter to the  $s$  direction as many times as there are unstable open-loop poles ( $N = -P$  in Eq. (7.27)). The proof of this criterion relies on Cauchy's principle of the argument and is given in most introductory textbooks on continuous control systems. The elementary interpretation is based on the following points:

- If we take values of  $s$  on a contour in the  $s$ -plane that encloses the unstable region, we can plot the corresponding values of the function  $1 + KD(s)G(s)$  in an image plane.
- If the  $s$ -plane contour encircles a *zero* of  $1 + KDG$  in a certain direction, the image contour will encircle the origin in the *same* direction. In the  $s$ -plane, the angle of the vector from the zero to  $s$  on the contour goes through  $360^\circ$ .
- If the  $s$ -plane contour encircles a *pole* of  $1 + KDG$ , the image contour will encircle the origin in the *opposite direction*. In this case, the  $s$ -plane vector angle also goes through  $360^\circ$  but the contribution to the image angle is a negative  $360^\circ$ .
- Thus the *net* number of same-direction encirclements,  $N$ , equals the difference  $N = Z - P$ .<sup>2</sup>
- The origin of the  $1 + KDG$  plane is the same as the point  $KDG = -1$  so we can plot  $KDG$  and count  $N$  as the encirclements of the  $-1$  point just as well.<sup>3</sup>

<sup>2</sup> It is much easier to remember same-direction and opposite-direction encirclements than to keep clockwise and counter-clockwise distinguished.

<sup>3</sup> When the characteristic equation is written as  $1 + KDG$ , we can plot only  $DG$  and count encirclements of  $DG = -\frac{1}{K}$  and thus easily consider the effects of  $K$  on stability and stability margins.



- From all of this, Eq. (7.27) follows immediately.

For the discrete case, the ideas are identical; the only difference is that the unstable region of the  $z$ -plane is the *outside* of the unit circle and it is awkward to visualize a contour that encloses this region. The problem can be avoided by the simple device of considering the encirclement of the *stable* region and calculating the stability result from that. The characteristic equation of the discrete system is written as

$$1 + K D(z)G(z) = 0, \quad (7.28)$$

and, as in the continuous case, it is assumed that the number,  $P$ , of unstable poles of  $K D(z)G(z)$ , which are also unstable poles of  $1 + K D(z)G(z)$ , is known and we wish to determine the number,  $Z$ , of unstable zeros of Eq. (7.28), which are the unstable closed-loop poles. Examination of Eq. (7.28) reveals that the (possibly unknown) total number of stable plus unstable poles,  $n$ , is the same as the total number of zeros of Eq. (7.28). Thus the number of *stable* zeros is  $n - Z$  and the number of *stable* poles is  $n - P$ . Following the mapping result used by Nyquist, the map of  $1 + K D(z)G(z)$  for the  $z$  contour of Fig. 7.18(b) will encircle the origin  $N$  times where

$$\begin{aligned} N &= \{\text{number of stable zeros}\} - \{\text{number of stable poles}\} \\ &= \{n - Z\} - \{n - P\} \\ &= P - Z. \end{aligned}$$

Therefore, the Nyquist stability criterion for discrete systems is

$$Z = P - N. \quad (7.29)$$

In summary, the discrete Nyquist stability criterion is

- Determine the number,  $P$ , of unstable poles of  $K D G$ .
- Plot  $K D(z)G(z)$  for the unit circle,  $z = e^{j\omega T}$  and  $0 \leq \omega T \leq 2\pi$ . This is a counter-clockwise path around the unit circle. Points for the plot can be conveniently taken from a discrete Bode plot of  $K D G$ .
- Set  $N$  equal to the net number of counter-clockwise (same direction) encirclements of the point  $-1$  on the plot.
- Compute  $Z = P - N$ . The system is stable if and only if  $Z = 0$ .

---

#### ◆ Example 7.9 Nyquist Stability

Evaluate the stability of the unity feedback discrete system with the plant transfer function

$$G(s) = \frac{1}{s(s+1)}, \quad (7.30)$$

with sampling at the rate of 1/2 Hz or  $T = 2$  and zero-order hold. The controller is proportional discrete feedback [ $KD(z) = K$ ].

**Solution.** The discrete transfer function at the specified sampling rate and ZOH is given by sysd3 of Example 7.8 with transfer function

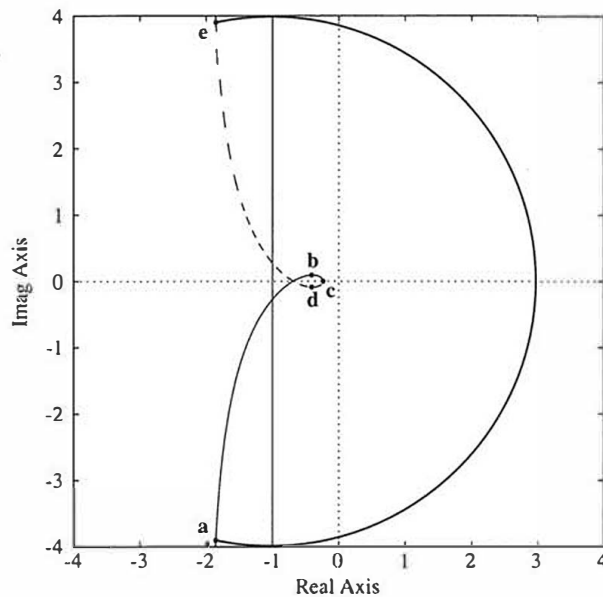
$$G(z) = \frac{1.135(z + 0.523)}{(z - 1)(z - 0.135)}, \quad (7.31)$$

and the plot of magnitude and phase of  $G(z)$  for  $z = e^{j\omega T}$  is included in Fig. 7.16 for  $0 \leq \omega T \leq \pi$ . Using the data from Fig. 7.16 for  $T = 2$ , the plot of  $KD(z)G(z)$  can be drawn as shown in Fig. 7.19. The plot is marked with corresponding points from Fig. 7.18(b) to facilitate understanding the results. Note that the portion from  $a \rightarrow b \rightarrow c$  is directly from Fig. 7.16, and the section from  $c \rightarrow d \rightarrow e$  is the same information reflected about the real axis. The large semicircle from  $e \rightarrow a$  is the analytically drawn map of the small semicircle about  $z = 1$  drawn by letting  $(z - 1) = re^{j\phi}$  in Eq. (7.31) for  $r \ll 1$  and  $-\frac{\pi}{2} \leq \phi \leq \frac{\pi}{2}$ . Because this system is open-loop stable and there are no  $-1$  point encirclements, we conclude that the closed-loop system will be stable as plotted for  $K = 1$ . Note that all the necessary information to determine stability is contained in the Bode plot information from Fig. 7.16, which determines the portion from  $a \rightarrow c$  in Fig. 7.19. Using MATLAB, the plot can be made by the statements

```
nyquist(sysd3)
axis equal
grid
```

The axis statement sets the x and y axes to have equal increments.

**Figure 7.19**  
Nyquist plot of  
Example 7.9



## 7.4.2 Design Specifications in the Frequency Domain

### Gain and Phase Margins

gain margin

phase margin

The Nyquist plot shows the number of encirclements and thus the stability of the closed-loop system. Gain and phase margins are defined so as to provide a two-point measure of how close the Nyquist plot is to encircling the  $-1$  point, and they are identical to the definitions developed for continuous systems. **Gain margin (GM)** is the factor by which the gain can be increased before causing the system to go unstable, and is usually the inverse of the magnitude of  $D(z)G(z)$  when its phase is  $180^\circ$ . The **phase margin (PM)** is the difference between  $-180^\circ$  and the phase of  $D(z)G(z)$  when its amplitude is 1. The PM is a measure of how much additional phase lag or time delay can be tolerated in the loop before instability results.

#### ◆ Example 7.10 Stability Margins

Consider the open-loop transfer function

$$G(s) = \frac{1}{s(s+1)^2},$$

with ZOH and sample rate of 5 Hz. The discrete transfer function is given by

$$G(z) = 0.0012 \frac{(z+3.38)(z+0.242)}{(z-1)(z-0.8187)^2}.$$

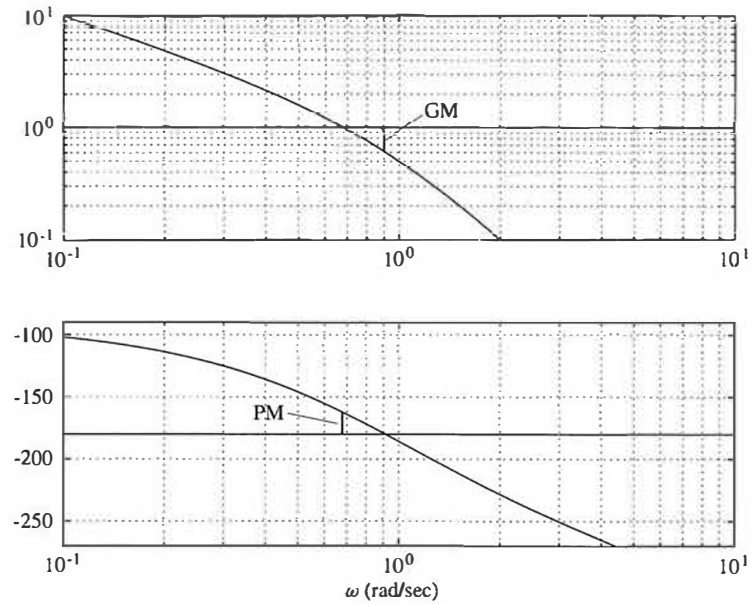
What are the gain and phase margins when in a loop with proportional discrete feedback ( $D(z) = K = 1$ )?

**Solution.** The discrete Bode plot is given in Fig. 7.20 and the portion of the Nyquist plot representing the frequency response in the vicinity of  $-1$  is plotted in Fig. 7.21. Unlike Example 13.5 which had a very slow sample rate, the higher sample rate here causes the magnitude to be essentially zero at  $\omega T = \pi$ , and hence the Nyquist plot goes almost to the origin. The plot is very similar to what would result for a continuous controller. Furthermore, just as in the continuous case, there are no  $-1$  point encirclements if  $K = 1$  as plotted ( $N = 0$ ), and since there are no unstable poles ( $P = 0$ ), the system will be stable at this gain ( $Z = 0$ ). If the Nyquist plot is multiplied by 1.8, then the plot will go through the  $-1$  point. Thus the gain margin is  $GM = 1.8$ . For values of  $K > 1.8$ , the  $-1$  point lies within the contour thus creating two encirclements ( $N = 2$ ) and two unstable closed-loop poles ( $Z = 2$ ). As indicated on the plot, the angle of the plot when the gain is 1 is  $18^\circ$  from the negative axis, so the phase margin is  $18^\circ$ .

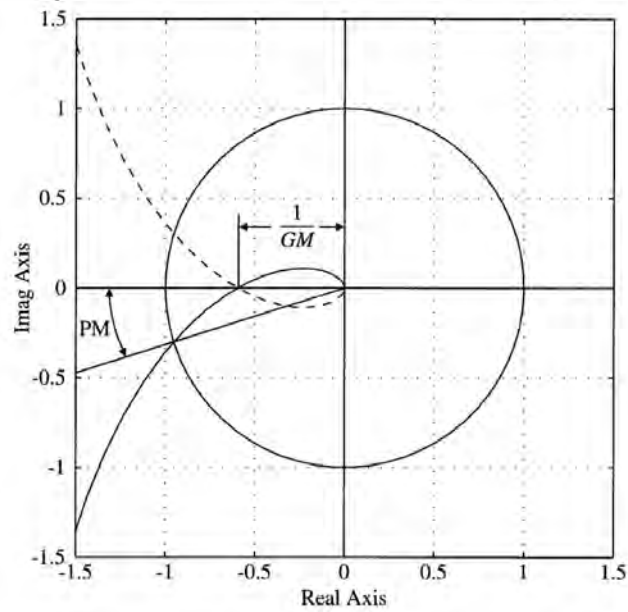
phase margin and  
damping ratio

For continuous systems, it has been observed that the phase margin,  $PM$ , is related to the damping ratio,  $\zeta$ , for a second-order system by the approximate

**Figure 7.20**  
Gain and phase margins  
on a Bode plot for  
Example 7.10

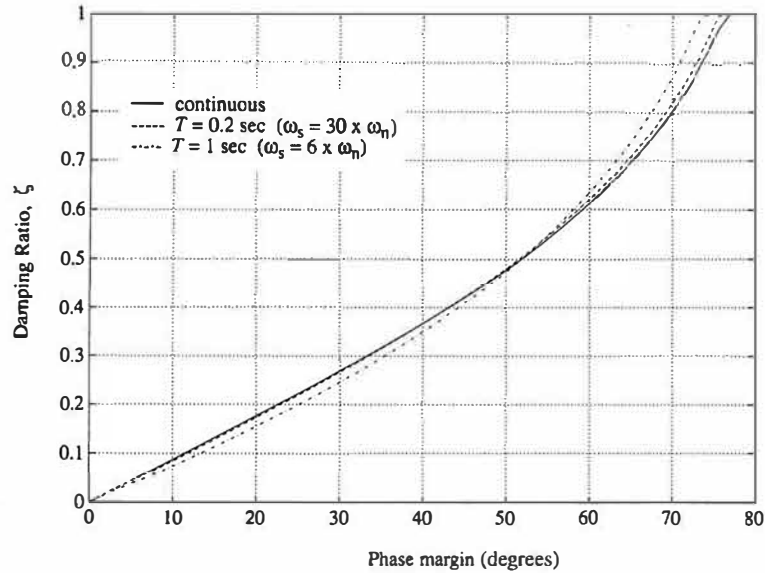


**Figure 7.21**  
Gain and phase margins  
on a Nyquist plot for  
Example 7.10



relation,  $\zeta \cong PM/100$ . This relationship is examined in Fig. 7.22 for the continuous case and for discrete systems with two values of the sample rate. Figure 7.22 was generated by evaluating the damping ratio of the closed-loop system

**Figure 7.22**  
Damping ratio of a  
second-order system  
versus phase margin  
( $PM$ )



that resulted when discrete proportional feedback was used with the open-loop system

$$G(s) = \frac{1}{s(s+1)}$$

A  $z$ -transform analysis of this system resulted in  $z$ -plane roots that were then transformed back to the  $s$ -plane via the inverse of  $z = e^{sT}$ . The  $\zeta$  of the resulting  $s$ -plane roots are plotted in the figure. As the feedback gain was varied, the damping ratio and phase margin were related as shown in Fig. 7.22. The actual sample rates used in the figure are 1 Hz and 5 Hz, which represent 6 and 30 times the open-loop system pole at 1 rad/sec. The conclusion to be drawn from Fig. 7.22 is that the  $PM$  from a discrete  $z$ -plane frequency response analysis carries essentially the same implications about the damping ratio of the closed-loop system as it does for continuous systems. For second-order systems without zeros, the relationship between  $\zeta$  and  $PM$  in the figure shows that the approximation of  $\zeta \cong PM/100$  is equally valid for continuous and discrete systems with reasonably fast sampling. For higher-order systems, the damping of the individual modes needs to be determined using other methods.

### Tracking Error in Terms of the Sensitivity Function

The gain and phase margins give useful information about the relative stability of nominal systems but can be very misleading as guides to the design of realistic

control problems. A more accurate margin can be given in terms of the sensitivity function. For the unity feedback system drawn in Fig. 7.1, the error is given by

$$E(j\omega) = \frac{1}{1 + DG} R \triangleq S(j\omega)R, \quad (7.32)$$

sensitivity function

where we have defined the **sensitivity function**  $S$ . In addition to being a factor of the system error, the sensitivity function is also the reciprocal of the distance of the Nyquist curve,  $DG$ , from the critical point  $-1$ . A large value for  $S$  indicates a Nyquist plot that comes close to the point of instability. The maximum value of  $|S|$  is often a more accurate measure of stability margin than either gain or phase margin alone. For example, in Fig. 7.23 a Nyquist plot is sketched that is much closer to instability than either gain or phase margin would indicate. The **vector gain margin** (VGM) is defined as the gain margin in the direction of the worst possible phase. For example, if the Nyquist plot comes closest to  $-1$  on the negative real axis, then the vector margin is the same as the standard gain margin. From the geometry of the Nyquist plot, the distance from the curve to  $-1$  is  $1 + DG = \frac{1}{S}$  and with the definition that

vector gain margin

$$S_{\infty} = \max_{\omega} |S|,$$

it follows that the distance of the closest point on the Nyquist curve from  $-1$  is  $\frac{1}{S_{\infty}}$ . If the Nyquist curve came this close to the  $-1$  point on the real axis, it would pass through  $1 - \frac{1}{S_{\infty}}$  and by definition, the product  $VGM \times (1 - \frac{1}{S_{\infty}}) = 1$ . Therefore we have that

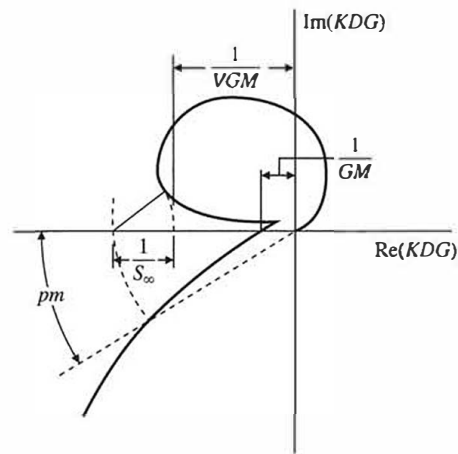
$$VGM = \frac{S_{\infty}}{S_{\infty} - 1}. \quad (7.33)$$

The VGM and related geometry are marked on the Nyquist plot in Fig. 7.23.

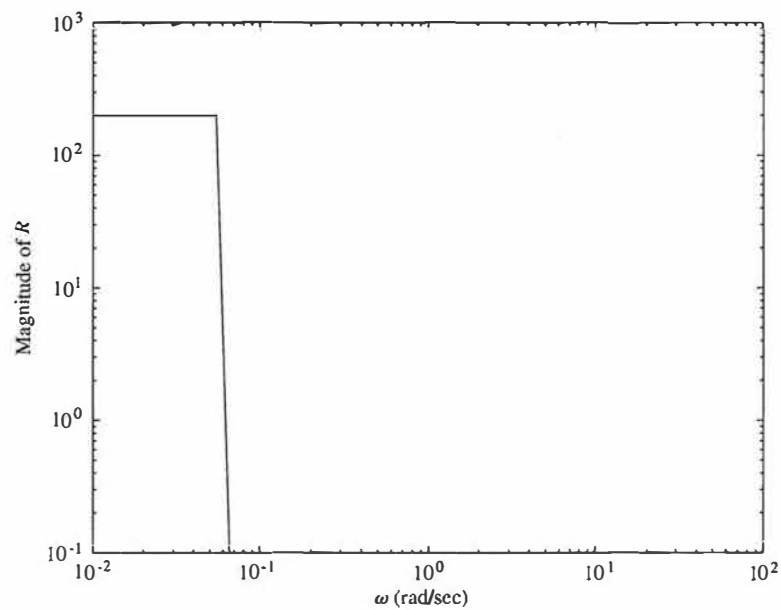
We can express more complete frequency domain design specifications than any of these margins if we first give frequency descriptions for the external reference and disturbance signals. For example, we have described so far dynamic performance by the transient response to simple steps and ramps. A more realistic description of the actual complex input signals is to represent them as random processes with corresponding frequency spectra. A less sophisticated description which is adequate for our purposes is to assume that the signals can be represented as a sum of sinusoids with frequencies in a specified range. For example, we can usually describe the frequency content of the reference input as a sum of sinusoids with relative amplitudes given by a magnitude function  $|R|$  such as that plotted in Fig. 7.24, which represents a signal with sinusoidal components each having about the same amplitude of 150 up to some value  $\omega_1$  and very small amplitudes for frequencies above that. With this assumption the response specification can be expressed by a statement such as “the magnitude of the system error is to be less than the bound  $e_b$  (a value such as 0.01 that defines the required tracking

**Figure 7.23**

A Nyquist plot showing the vector gain margin

**Figure 7.24**

Sketch of typical specification of frequency content for reference input tracking



accuracy) for any sinusoid of frequency  $\omega_o$  and of amplitude given by  $|R(j\omega_o)|$ .” We can now define the size of the error in terms of the sensitivity function and the amplitude of the input. Using Eq. (7.32), the frequency-based error specification can be expressed as  $|E| = |S| |R| \leq e_b$ . In order to normalize the problem without

defining both the spectrum  $R$  and the error bound each time, we define the real function of frequency  $W_1(\omega) = |R|/e_b$  and the requirement can be written as

$$|S| W_1 \leq 1. \quad (7.34)$$

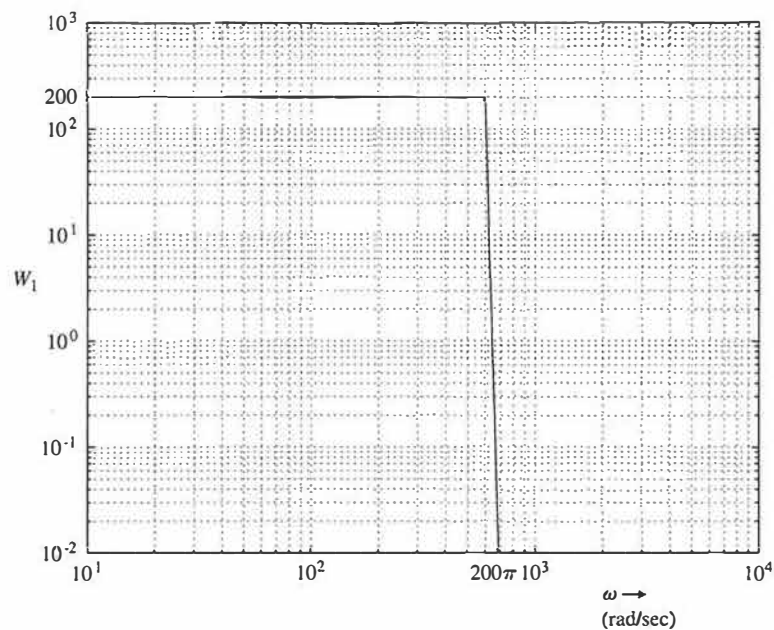
◆ **Example 7.11** *Performance Bound Function*

A unity feedback system is to have an error less than 0.005 for all unity amplitude sinusoids having frequency below 100 Hz. Draw the performance frequency function  $W_1(\omega)$  for this design.

**Solution.** The spectrum, from the problem description, is unity for  $0 \leq \omega \leq 200\pi$ . Since  $e_b = 0.005$ , the required function is given by a rectangle of amplitude  $1/0.005 = 200$  over the given range. The function is plotted in Fig. 7.25.

The expression in Eq. (7.34) can be translated to the more familiar Bode plot coordinates and given as a requirement on the open-loop gain,  $DG$ , by observing

**Figure 7.25**  
Plot of performance  
frequency function for  
Example 7.11





that over the frequency range when errors are small the loop gain is large. In that case  $|S| \approx \frac{1}{|DG|}$  and the requirement is approximately

$$\begin{aligned} \frac{W_1}{|DG|} &\leq 1 \\ |DG| &\geq W_1. \end{aligned} \quad (7.35)$$

### Stability Robustness in Terms of the Sensitivity Function

In addition to the requirements on dynamic performance, the designer is usually required to design for stability robustness. The models used for design are almost always only approximations to the real system. Many small effects are omitted, such as slight flexibility in structural members or parasitic electrical elements in an electronic circuit. Usually these effects influence the transfer function at frequencies above the control bandwidth and a nominal transfer function,  $G_o$ , is used for the design. However, while the design is done for the nominal plant transfer function, the actual system is expected to be stable for an entire class of transfer functions that represent the range of changes that are expected to be faced as all elements are included and as changes due to temperature, age, and other environmental factors vary the plant dynamics from the nominal case. A realistic way to express plant uncertainty is to describe the plant transfer function as having a multiplicative uncertainty as

$$G(j\omega) = G_o(j\omega)[1 + w_2(\omega)\Delta(j\omega)]. \quad (7.36)$$

In Eq. (7.36),  $G_o(j\omega)$  is the nominal plant transfer function, and the real function,  $w_2(\omega)$ , is a magnitude function that expresses the size of changes as a function of frequency that the transfer function is expected to experience and is *known to be less than some upper bound*  $W_2(\omega)$ . The value of the bound  $W_2$  is almost always very small for low frequencies (we know the model very well there) and increases substantially as we go to high frequencies where parasitic parameters come into play and unmodeled structural flexibility is common.

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#### ◆ Example 7.12 Model Uncertainty

A magnetic memory read/write head assembly can be well modelled at low frequencies as

$$G_o(s) = \frac{K}{s^2}. \quad (7.37)$$

However, the arm supporting the read/write head has some lightly damped flexibility with uncertain resonant frequency. With scaling to place the resonant frequency at  $\omega_o$ , and damping  $B$ , the more accurate model is represented as

$$G(s) = \frac{K}{s^2} \frac{B \frac{s}{\omega_o} + 1}{\left(\frac{s}{\omega_o}\right)^2 + B \frac{s}{\omega_o} + 1} \quad (7.38)$$

Compute the model uncertainty function for this case.

**Solution.** The model transfer function given by Eq. (7.38) can be written as

$$G(s) = \frac{K}{s^2} \left[ 1 + \frac{-\left(\frac{s}{\omega_o}\right)^2}{\left(\frac{s}{\omega_o}\right)^2 + B \frac{s}{\omega_o} + 1} \right] \quad (7.39)$$

Comparing Eq. (7.39) with Eq. (7.36), the model uncertainty function is given by

$$w_2(\omega) = \left| \frac{-\left(\frac{s}{\omega_o}\right)^2}{\left(\frac{s}{\omega_o}\right)^2 + B \frac{s}{\omega_o} + 1} \right|_{s=j\omega} \quad (7.40)$$

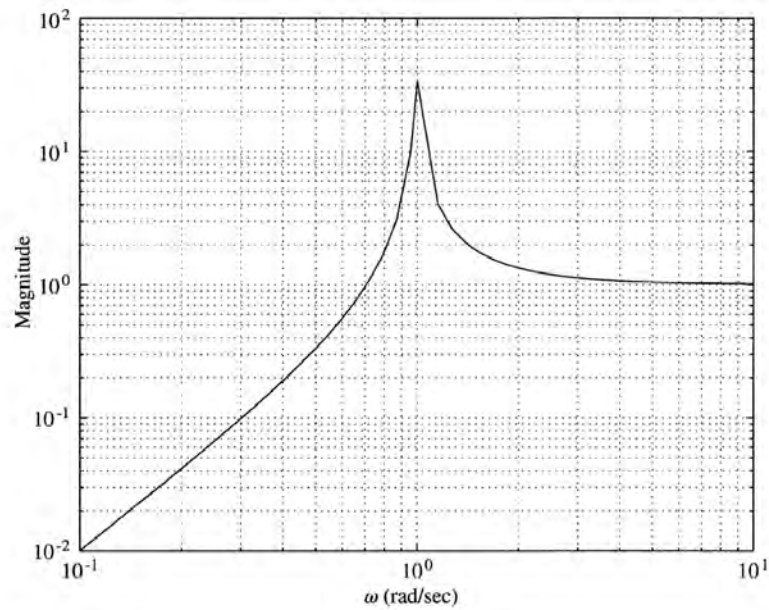
A plot of this function is given in Fig. 7.26 for  $\omega_o = 1$  and  $B = .03$ .

In general, a model uncertainty bound is small for low frequencies and large for higher frequencies. A typical shape is sketched in Fig. 7.27. The complex function,  $\Delta(j\omega)$ , represents the uncertainty in phase and is restricted only by the constraint

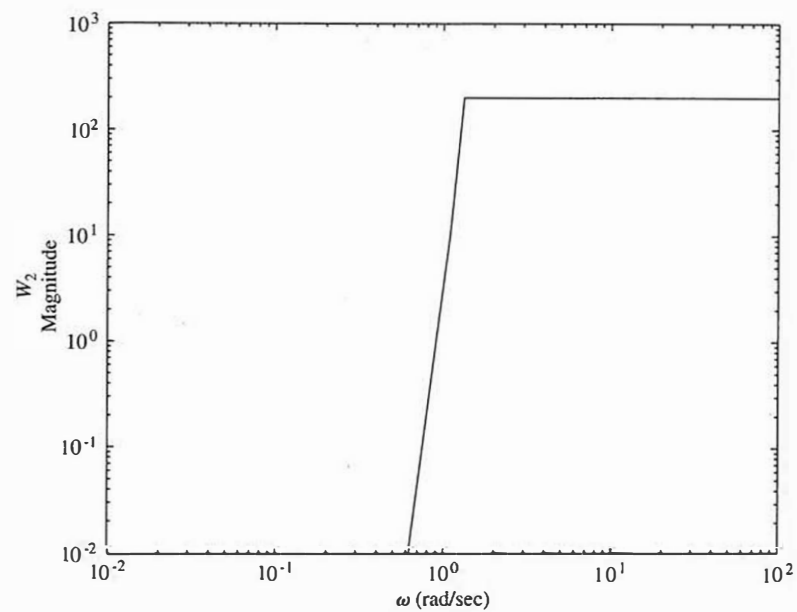
$$|\Delta(j\omega)| \leq 1. \quad (7.41)$$

Stability robustness requires that we construct a control design for  $G_o(s)$  which will result in a stable system for any transfer function described by Eq. (7.36). To derive the requirement, we begin with the assumption that the nominal design has been done and is stable so that the Nyquist plot of  $DG_o$  satisfies the Nyquist stability criterion. In this case, the equation  $1 + D(j\omega)G_o(j\omega) = 0$  is never satisfied for any real frequency. If the system is to have stability robustness, the characteristic equation using the uncertain plant as described by Eq. (7.36)

**Figure 7.26**  
Plot of model  
uncertainty function for  
disk drive read/write  
head assembly



**Figure 7.27**  
A plot of a typical plant  
uncertainty frequency  
function



must not go to zero for any real frequency for any value of either  $W_2$  or  $\Delta$ . The requirement can be written as a function of  $j\omega$  in the form

$$1 + DG \neq 0, \quad (7.42)$$

$$1 + DG_o[1 + w_2\Delta] \neq 0, \quad (7.43)$$

$$(1 + DG_o) \left( 1 + \frac{DG_o}{1 + DG_o} w_2\Delta \right) \neq 0, \quad (7.44)$$

complementary  
sensitivity function

where the **complementary sensitivity function** is defined as  $T(j\omega) \triangleq DG_o / (1 + DG_o) = 1 - S$ . Because the nominal system is stable, the first term in Eq. (7.42),  $(1 + D(j\omega)G_o(j\omega))$ , is not zero for any  $\omega$ . Thus, if Eq. (7.42) is not to be zero for any frequency, any  $w_2 \leq W_2$ , or for any phase function  $\Delta$ , then it is necessary and sufficient that

$$\begin{aligned} |Tw_2\Delta| &< 1, \\ |T| |w_2| |\Delta| &< 1, \end{aligned}$$

which reduces to

$$|T| W_2 < 1, \quad (7.45)$$

making use of Eq. (7.41) and the fact that  $w_2$  is bounded by  $W_2$ . As with the performance specification, for single-input-single-output unity feedback systems this requirement can be approximated by a more convenient form. Over the range of high frequencies where there is significant model uncertainty and  $W_2$  is non-negligible,  $DG_o$  is small. Therefore we can approximate  $T \approx DG_o$  and the constraint becomes

$$\begin{aligned} |DG_o| W_2 &< 1 \\ |DG_o| &< \frac{1}{W_2}. \end{aligned} \quad (7.46)$$

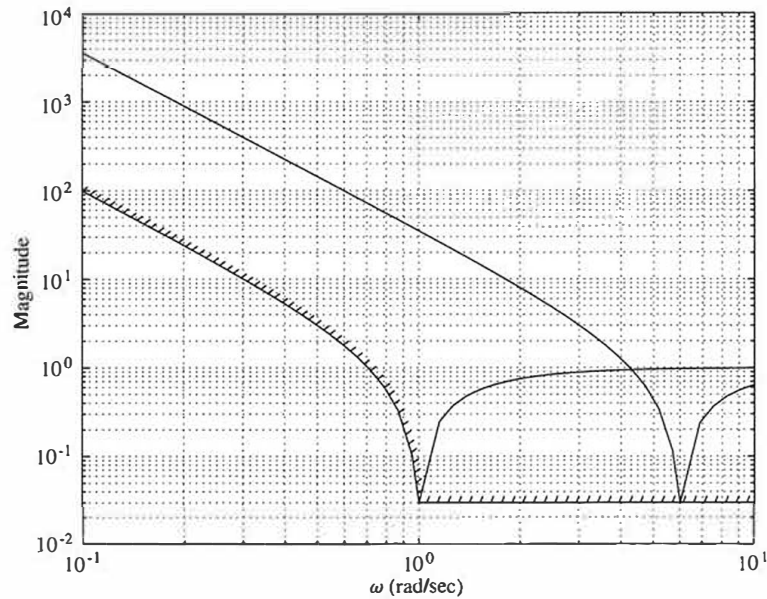
### ◆ Example 7.13 Stability Robustness Function

The uncertainty in the model of a disk read/write head assembly is given in Example 7.12. Suppose it is known that the parameter  $B$  is restricted to the range  $.03 \leq B \leq .3$  and that the resonant frequency  $\omega_o$  is known to be no less than 1.0. Plot the stability robustness bound,  $1/W_2$  for this problem.

**Solution.** The function  $1/w_2(\omega)$  is plotted for  $B = .03$  and  $\omega = 1$  and  $\omega = 6$  in Fig. 7.28 using bode. It is clear that if the resonant frequency can take on any value greater than 1.0, then the bound  $1/W_2$  needs to be extended at the value .03 for all frequencies greater than 1.0. The boundary line is marked with hatching in Fig. 7.28.

**Figure 7.28**

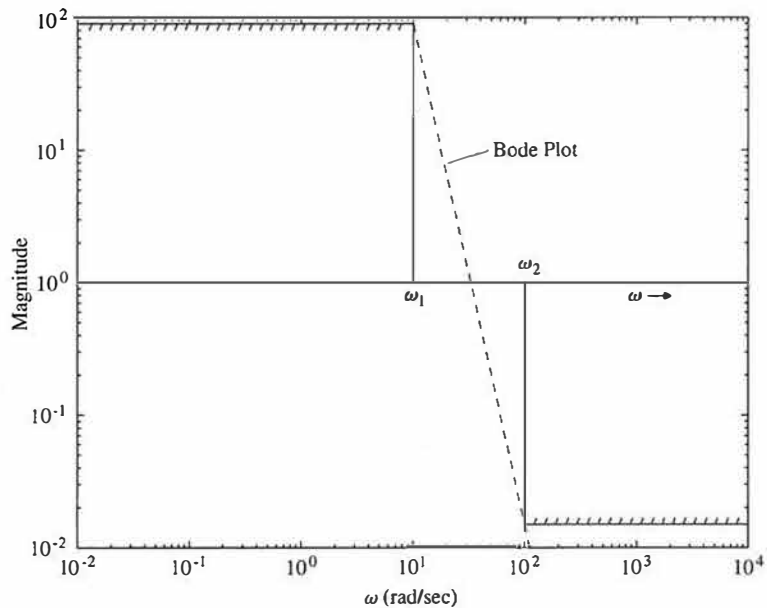
Plot of the stability robustness frequency function for disk read/write head assembly



In practice, the magnitude of the loop gain is plotted on log-log coordinates, and the constraints of Eq. (7.35) and Eq. (7.46) are included on the same plot. A

**Figure 7.29**

Typical design limitations as displayed on a Bode magnitude plot



typical sketch is drawn in Fig. 7.29. The designer is expected to construct a loop gain that will stay above  $W_1$  for frequencies below  $\omega_1$ , cross over the magnitude-of-1 line ( $\log(|DG|) = 0$ ) in the range  $\omega_1 \leq \omega \leq \omega_2$  and stay below  $1/W_2$  for frequencies above  $\omega_2$ . We have developed the design constraints Eq. (7.35) and Eq. (7.46) in terms of  $j\omega$  as for continuous systems. The algebra and the equations are the same for the discrete case; one need only substitute the discrete transfer functions for the continuous ones and use the variable  $e^{j\omega T}$ .

### Limitations on Design: Continuous Case

Bode's gain-phase  
formula

One of the major contributions of Bode was to derive important limitations on transfer functions that set limits on achievable design specifications. For example, we would like to have the system error kept small for the widest possible range of frequencies and yet to have a system that is stable in the presence of uncertainty in the plant transfer function. In terms of the plot in Fig. 7.29, we want  $W_1$  and  $W_2$  to be large in their respective frequency ranges and for  $\omega_1$  to be close to  $\omega_2$ . Thus the loop gain is expected to plunge with a large negative slope from being greater than  $W_1$  to being less than  $1/W_2$  in a short span, while maintaining stability which can be expressed as having a good phase margin. Bode showed that this is *impossible* with a linear controller by showing that the minimum possible phase is determined by an integral depending on the slope of the magnitude curve. A common form of the formula for phase is

Bode's gain-phase  
integral

$$\phi(\omega_o) = \frac{1}{\pi} \int_{-\infty}^{+\infty} \left( \frac{dM}{du} \right) \ln(\coth \left| \frac{u}{2} \right|) du, \quad (7.47)$$

where  $M = \ln(|DG|)$  and  $u = \ln(\frac{\omega}{\omega_o})$ , and thus  $\frac{dM}{du}$  is the magnitude slope on the log-log (Bode) plot. The weighting function in Eq. (7.47) is concentrated near  $\omega_o$ , and if the slope is constant for a substantial range around  $\omega_o$ , then the formula can be approximated by

$$\phi(\omega_o) \approx \frac{\pi}{2} \left. \frac{dM}{du} \right|_{u=0}.$$

If, for example, the phase is to be kept above  $-150^\circ$  to maintain a  $30^\circ$  phase margin, then the magnitude slope is estimated to be

$$\begin{aligned} \frac{dM}{du} &\approx \frac{2}{\pi} \left( -150 \frac{\pi}{180} \right) \\ &\approx -1.667 \end{aligned}$$

in the neighborhood of crossover. If we try to make the average slope steeper (more negative) than this near crossover, we will lose the phase margin. From this condition there developed the design rule:

- The asymptotes of the Bode plot magnitude, which are restricted to be integral values for rational functions, should be made to crossover the magnitude 1

line at a slope of  $-1$  over a frequency range of about one decade around the cross-over frequency.

Modifications to this rule need of course to be made in particular cases, but the limitation expressed by Eq. (7.47) is a hard limit that cannot be avoided.

sensitivity integral

In Freudenberg and Looze (1985) an extension to another of Bode's relations was derived. This is a constraint on the integral of the sensitivity function dependent on the presence of open-loop right-half plane poles. Suppose the loop gain  $DG_o$  has  $n_p$  poles,  $p_i$ , in the right-half plane and "rolls off" at high frequencies at a slope faster than  $-1$ . For rational functions, this means that there is an excess of at least two more total poles than zeros. Then it can be shown that

$$\int_0^\infty \ln(|S|) d\omega = \pi \sum_{i=1}^{n_p} \operatorname{Re}\{p_i\}. \quad (7.48)$$

If there are no right-half plane poles, then the integral is zero. This means that if we make the log of the sensitivity function very negative over some frequency band to reduce errors in that band, then *of necessity*  $\ln |S|$  will be positive over another part of the band and errors will be amplified there. If there are unstable poles, the situation is worse because the positive area where sensitivity magnifies the error must exceed the negative area where the error is reduced by the feedback. There are also consequences if  $DG_o$  has any *zeros* in the right-half plane. If the open-loop system has no zeros in the right-half plane, then it is in principle possible to keep the magnitude of the sensitivity small by spreading the sensitivity increase over all positive frequencies to infinity but such a design requires an excessive bandwidth and is rarely practical. If a specific bandwidth is imposed, then the sensitivity function is constrained to take on a finite, possibly large, positive value at some point below the bandwidth and a large value of  $|S|_\infty$  leads to a small VGM and generally an unsatisfactory design.

constraint on  
non-minimum phase  
systems

An alternative to Eq. (7.48) is also true if there is a (non-minimum-phase) zero of  $DG_o$  in the right-half plane. Suppose the zero is located at  $z_o = \sigma_o + j\omega_o$ , where  $\sigma_o > 0$ . Again we assume there are  $n_p$  right-half plane poles at locations  $p_i$  with conjugate values  $\bar{p}_i$ . Now the condition can be expressed as a two-sided weighted integral

$$\int_{-\infty}^{\infty} \ln(|S|) \frac{\sigma_o}{\sigma_o^2 + (\omega - \omega_o)^2} d\omega = \pi \sum_{i=1}^{n_p} \ln \left| \frac{\bar{p}_i + z_o}{p_i - z_o} \right|. \quad (7.49)$$

In this case, we do not have the "roll-off" restriction and there is no possibility of spreading the positive area over all high frequencies because the weighting function goes to zero with frequency. The important point about this integral is that if the non-minimum phase zero is close to a right-half plane pole, the right side of the integral can be very large and the excess of positive area is required to be correspondingly large. Based on this result, *one expects especially great*

*difficulty meeting specifications on sensitivity with a system having right-half plane poles and zeros close together.*

### Limitations on Design: Discrete Case

In the discrete case, the relation between gain slope and phase does not hold although it is approximately true for frequencies well below the Nyquist frequency. The situation is illustrated by the Bode plot in Fig. 7.30 for the plant  $G = 1/s^2$ . Notice that the phase is always slightly more negative than the  $-180^\circ$  one would get for this plant in the continuous case and deviates more as we approach the Nyquist limit at  $\pi/T$ . The effect is approximated by the delay of  $T/2$  due to the sample and ZOH. From this example, one suspects that the restriction on phase due to gain slope is *more severe* in the discrete case than in the continuous case.

In Sung and Hara (1988) the discrete versions of these design limitations are derived. We consider the single-loop unity feedback structure and define the sensitivity function as  $S = \frac{1}{1+D(z)G(z)}$  as before. We also assume that the open-loop transfer function  $D(z)G(z)$  has  $n_p$  poles outside the unit circle at  $z_i = r_i e^{j\phi_i}$ ,  $r_i > 1$ . It can be shown that

$$\int_0^\pi \ln(|S(e^{j\phi})|) d\phi = \pi \sum_{i=1}^{n_p} \ln(r_i). \quad (7.50)$$

The implications of Eq. (7.50) are the same as in the continuous case except for the fact that the integral in Eq. (7.50) is over a *finite* limit. If we require that

discrete sensitivity  
integral

**Figure 7.30**  
Discrete Bode plot of  
 $1/s^2$  plant with  
zero-order hold

